

Weekly Update — NSF Funding Analysis

1. What did you accomplish this week?

I finished cleaning the data to 2021the 2025. Also redid the analysis on LDA and bertopics with better topics/words for each categories. The graphs for the top 10 and top 30s also change to be able to see all. Used last week's comments to change to treemap for graphs that are not so great as bar graphs.

2. Did you learn or discover anything new?

The most surprising finding is that nearly every cs subfield is declining in both awards and funding from 2021 to 2025, except Scientific Computing, Cyberinfrastructure, and Deep Learning. The biggest declines are in Networking, HCI, K-12 Education, and Computer Vision. This suggests NSF is consolidating funding around AI/ML and computing infrastructure while pulling back from other areas. The corpus model also picked up topics the abstract model missed — specifically Bioinformatics, Graph Neural Networks, and Large Language Models — because award titles contain more specific technical terms.

3. Are you uncertain, stuck, or worried about anything?

The interactive charts don't render in the pdf, so some charts appear blank. I worked around this by exporting to HTML first and printing from Safari, but the Plotly charts are still missing. I resolved this by just removing interactive charts completely. I'm also not 100% sure my topic-to-subfield label mappings are correct — some topics had overlapping themes.

4. What is your plan for next week? Write up the findings and interpretation, clean up the notebook with markdown explanations for each section, and finalize the pdf.

5. Anything else? No, I think I'm on track overall. The core analysis is complete.